INVESTIGATING THE RELATIONSHIP BETWEEN FOCAL SEIZURE DURATION AND LOCALIZATION USING R:

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Abstract:

In the United States, approximately 150,000 individuals are diagnosed with epilepsy every year. Nearly 60% of epileptic seizures are focal seizures, meaning they begin in one area of the brain. Understanding how they work is critical to advancing medical treatment.

Current therapies for epilepsy include anti-seizure medications, dietary therapies, surgical interventions, and neurostimulation devices. However, success for many of these methods depends on understanding where seizures begin and how they behave. For focal seizures, localization informs electrode placement for stimulation, determines which section of the brain will be surgically removed, and helps predict seizure outcomes. To gain deeper insight, studying the relationship between seizure duration and onset location can reveal how specific neural networks and anatomical areas influence seizure initiation and termination.

Previous studies have explored seizure localization at the hemispheric level and its effect on seizure duration (Seethaler et al., 2021), but this project explores that relationship on a finer level by investigating how precise electrode sites impact variations in seizure length. Through line length and wavelet techniques, 64 seizures from the CHB-MIT EEG Scalp Database were analyzed, of which 44 were focal seizures.

The results consisted of seizures originating from many different channels, yet several distinct patterns still emerged. This suggests the possibility of predicting seizure duration based on their origin in certain channels. Further investigation with larger datasets should help identify even more nuanced connections between exact seizure onset locations and their durations, ultimately guiding the development of new therapies and improving existing ones.

Background Information:

- Around 2.3 million adults and 500,000 children live with some form of epilepsy (National Institute of Neurological Disorders and Stroke).
- This neurological disorder is characterized by recurring seizures which are caused by disturbances in neuronal activity.
- Though treatments like medication and surgery exist, about 30% of individuals have treatment-resistant epilepsy. This means they will keep experiencing seizures even with the best treatment available.
- Electroencephalography (EEG) offers a non-invasive method to track electrical activity across multiple brain regions. This allows researchers to identify channels associated with seizure onset.
- Mapping the relationship between electrode sites and seizure timing may clarify epileptic network mechanisms and guide treatment for complex cases.

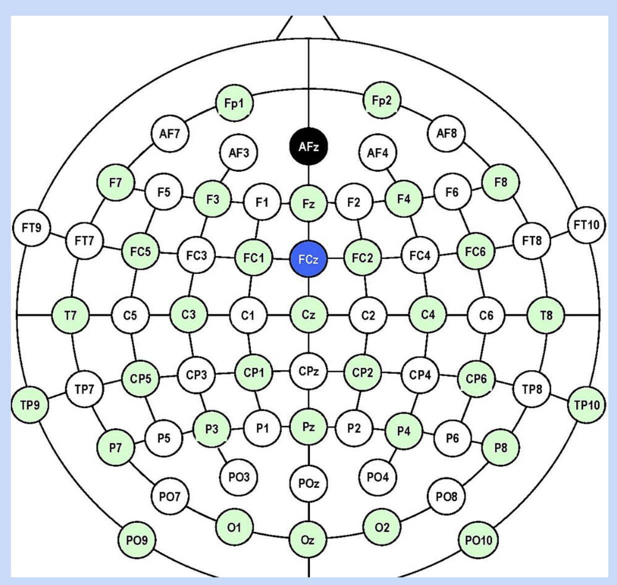

Figure 1: Figure shows surface map of eeg electrode locations. (Rupasov et al., 2012)

Method/Materials:

- **CHB-MIT Scalp EEG Database**
  - Contains a collection of EEG recordings from 22 patients with intractable seizures
  - Utilized 64 of these recordings as data points
- **Python (v.3.13) Miniconda**
  - Transformed files from edf to csv for compatible processing
  - Extracted 60 seconds pre-ictal period + seizure segment + 20 seconds postictal period from each eeg recording
- **RStudio (v.4.5.1)**
  - Filter out slow drifts and high frequency noise using Butterworth band-pass filter (1-70 Hz)
  - Conduct line length analysis to determine onset zone
  - Conduct wavelet analysis to determine onset zone/track seizure spread
  - Plotting heatmaps for both line length and wavelet analysis

Results:

- Seizures **chb17_04** and **chb23_06** involved the same bipolar channels **(FP1–F3 and FP2–F4)** in localization, and both had **durations between 110 and 115 seconds**. Seizure durations differed by 2 seconds in length.
- When bipolar channel **FT9.FT10** is the channel detected earliest for onset (but not the only one involved), seizure **durations are always under 35 seconds. Chb06_18** was 12 seconds, **chb06_24** was 16 seconds, and **chb15_10** was 31 seconds.
- Additionally, when bipolar channel **FZ.CZ** is the channel detected earliest for onset (by at least half a second), seizure **durations are generally longer than 75 seconds**. **Chb04_08** was 111 seconds long, **chb05_22** was 117 seconds long, **chb09_08** was 79 seconds long, and **chb11_99** was 752 seconds long.

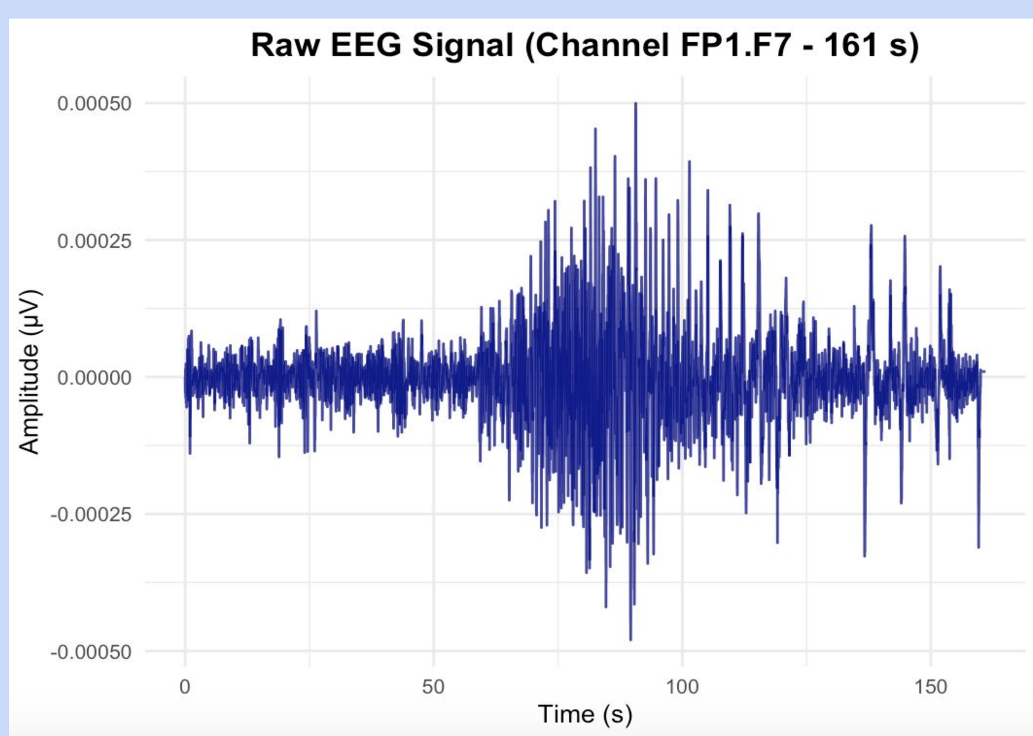

Figure 2: Plot of raw eeg signal from recording chb1_03, highlighting the transition from baseline (60 s pre-seizure) to the seizure event and recovery phase (40 s post-seizure) in the bipolar channel FP1–F7.

Results (continue):

- When bipolar channel **C4.P4** is the channel detected earliest for onset, seizure **durations are between 80 and 90 seconds**. **Chb02_16** is 82 seconds long, **chb07_12** is 86 seconds long, **chb10_38** is 89 seconds long.
- When **P8.O2** is the earliest channel to be detected for onset, seizure **durations are between 5 and 20 seconds. Chb02_19** and **chb16_17** each lasted 9 seconds, and **chb23_08** lasted 20 seconds.

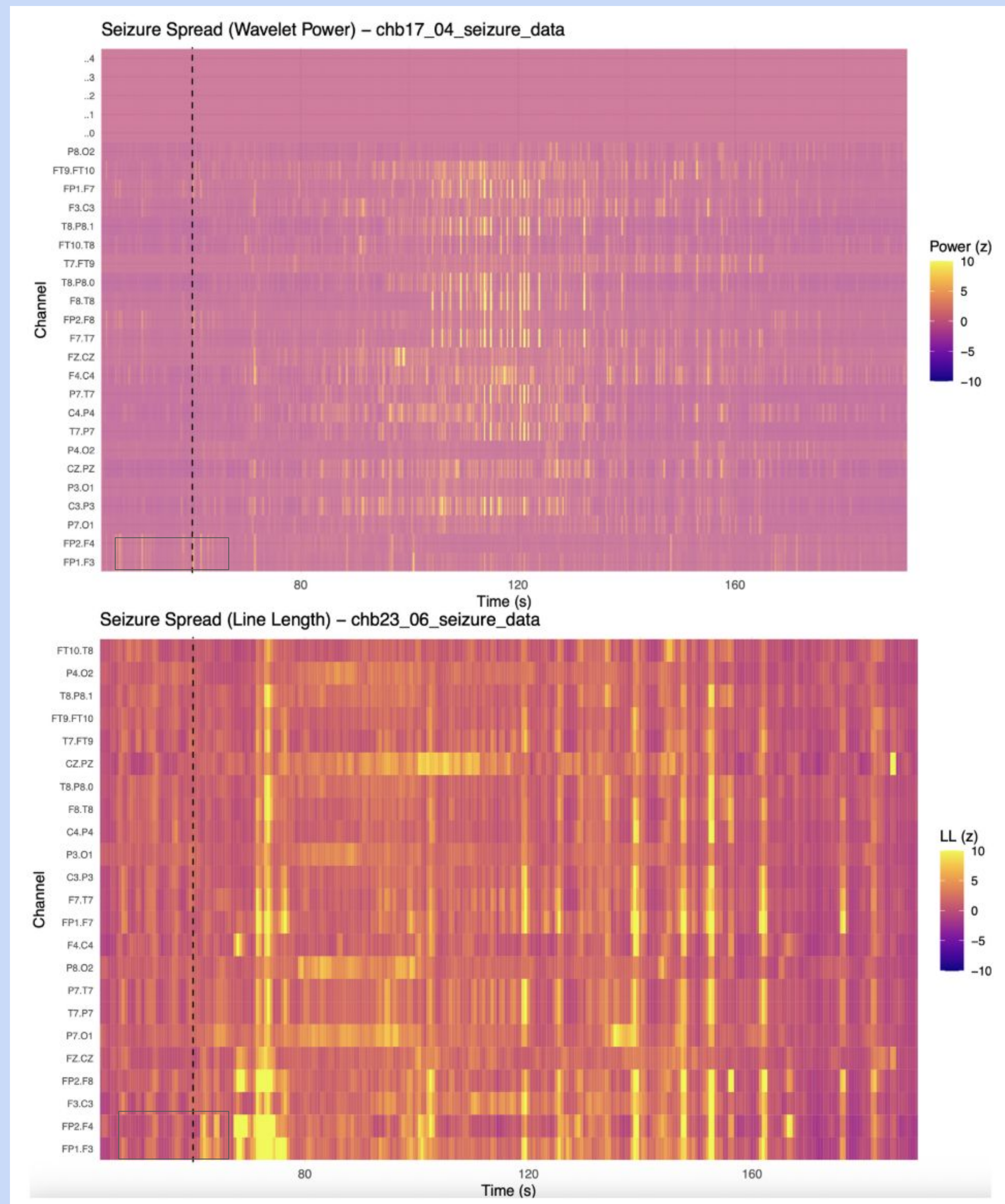

Figure 4: Heatmaps show early onset in channels FP1-F3 and FP2-F4 for seizures chb17_04 and chb23_06.

Discussion:

- Did not observe any consistent relationship between seizure duration and hemisphere/other brain region. However, for female patients, seizures that originated in the left hemisphere lasted longer on average when compared to the right hemisphere.
- Likewise, there was no distinct correlation between seizure type (generalized vs focal) and duration. Both types showed large duration ranges.
- Findings for this study revealed that certain earliest-onset bipolar channels were associated with characteristic seizure duration ranges.
- These patterns suggest that seizure onset location at the resolution of bipolar channel pairs may influence propagation dynamics and overall seizure length.

Conclusion:

- Though results seem promising, there were some limitations to this study, a key one being the relatively small number of seizures analyzed. As only 44 focal seizures were detected and analyzed, it is possible that certain trends weren't represented accurately and some were missed altogether.
- Future work should expand the dataset to include a greater variety of seizure types, patient demographics, and electrode configurations to improve the robustness and generalizability of the findings. This could help clarify whether onset location-based duration trends hold across patient populations and seizure types.

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Acknowledgements:

I would like to thank Dr. Antonios E. Marsellos and the Geology, Environment and Sustainability department at Hofstra University for providing us with the necessary tools to conduct this research.

Source (Google Slides):

https://docs.google.com/presentation/d/1lRAxmTlOps8mNThkxa4uBL6Ag7_WumGQkUcYGB2SAac/edit

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